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Research Article

Designing of low band gap organic semiconductors through data mining from multiple databases and machine learning assisted property prediction

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ABSTRACT

Bandgap is a key parameter for selecting suitable materials for a broad range of applications. Organic solar cells (OSCs) are emerging as powerful devices due to their low-cost solution processing. Developing OSCs necessitates producing effective materials in a computationally cost-effective and rapid manner. Machine learning has become popular and well-recognized among researchers to screen and design high performance materials for OSCs. Machine learning models require data from the literature (reported studies or databases) to effectively predict targeted properties. To unveil the hidden dataset patterns, a thorough data visualization analysis is conducted. Importantly, multiple database mining is performed for designing low band gap organic semiconductors. Molecular descriptors are utilized to train machine learning models. Importantly, about 22 different machine learning models are tested. Among all models, extra trees regressor shows higher predictive capability. Residuals, learning curve and validation curve are also drawn for extra trees regressor. Feature importance analysis determines the significance of the features. Moreover, library enumeration and similarity analysis further facilitate designing of high-performance semiconductor materials. This work may help in screening and designing efficient semiconductors having low band gap for increasing the efficiency of OSCs.

1. Introduction

Organic solar cells (OSCs) are the third type of solar cell technology [1–3]. OSCs are one of the arising photovoltaic advancements that uses organic polymer material as light absorbing layer [4]. Organic photovoltaics (OPVs) are thin film-oriented cells that can store larger amounts of solar energy than other sun-driven innovations. Inherited properties such as charge transport and light absorption/retention capacity of the organic material are the key advantages in the development of OSCs [5–7]. Organic semiconductors are π-conjugated organic molecules comprised of carbon and hydrogen atoms along with some heteroatoms including sulfur, oxygen, and nitrogen [8–10]. These have efficient electrical conductivity and electroluminescence characteristics [11,12]. Metal complexes, polycyclic aromatic hydrocarbons, and other substances produced from conductive polymers, such as polythiophenes, polyphenylamines, and polyacenes are examples of organic semiconductor materials [13–15].

Particularly, band gap is a crucial parameter for controlling the photovoltaic properties of semiconductor materials [16–18]. Low band gap organic semiconductors require low energy to transfer electron from valence to conduction band [19,20]. A low band gap enables the semiconductor to have higher electron mobility from the valence to conduction band, enhancing their conductivity [21,22]. Moreover, the low band gap semiconductors can harvest more photon as compared to wide band gap semiconductors, enabling higher solar energy conversion efficiency. In high band gap semiconductors, higher energy is required for electron transfer, which can decrease the electrical conductivity. Therefore, low band gap organic semiconductors show immense

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potential in a wide range of applications [23,24].

To further expand the applicability of organic semiconductors, materials with improved properties are needed [25,26]. The applications of these low band gap organic semiconductors include organic field-effect transistors (OFETs), organic light-emitting diodes (OLEDs), OSCs, supercapacitors, thin film batteries, and so on [27,28]. Moreover, the industrial utilization of semiconductors is limited due to failure in yielding high charge carrier mobilities in a reproducible manner. Due to intense research efforts, a variety of materials have been developed with higher mobilities, which can even exceed amorphous silicon [29]. However, due to the existence of a large variety of organic building blocks, it is assumed that these few developed materials are nothing. By assembling potential organic building blocks, several new organic molecules can be designed.

To date, experimental approaches require a complete cycle for the discovery of new materials including synthetic procedures, manufacturing of device, evaluation of performance, molecular tuning, trial, and error experiences [30,31]. Computational/theoretical investigations can play a significant role in the advancement of OPVs by providing further insight into the electronic properties (charge-transport mechanisms) [32–34]. These investigations can be made further useful by utilizing these in combination with experiments [35–37]. In recent years, various groups have been working on the discovery of high-performance materials to guide experimentalists for synthesizing potential materials/targets [38,39].

Machine learning, a subfield of artificial intelligence, is emerging as a powerful approach for screening and designing potential targets for various applications including drug discovery, super hard materials, heterogeneous catalysis, with the aid of large-scale data mining approaches for extracting a general design criterion [40,41]. In these screening/designing approaches, easily calculable or readily listed properties of molecules (systems) known as molecular descriptors can be correlated with crucial properties of molecules/materials/systems (e.g., carrier mobilities, band gap) are determined. These strategies can help to screen potential candidates for targeted applications [42,43]. Although various efforts have been made to discover potential materials for diverse applications, only a limited number of approaches have been focused on organic semiconductors for photovoltaic applications [44,45]. By utilizing the potential advantages of machine learning approaches, the screening of multiple databases can be rapidly achieved with highest accuracy and computationally low-cost manner [46].

In this work, machine learning is used for designing novel semiconductors with low band gap and property prediction for OSCs. For this purpose, data is collected. Different regression models are tested and the best one, i.e., extra trees regressor, is selected. To support the final model cook's distance, *t*-SNE, learning curve and validation curve is drawn. Structure based similarity analysis and library enumeration is done to provide useful information regarding semiconductor molecules. This study will help researchers in designing potential candidates with improved performance for photovoltaics.

2. Computational procedures

2.1. Machine learning analysis

Data of 300 semiconductors is collected from literature. Band gap (target property) is taken in eV. Data is provided in Table S1. Cook's distance provides an estimate of instances' influence on linear regression. It is considered as a primary parameter to classify the points negatively impacting the regression models. To identify different outliers in array of predicting variables, cook's distance is employed in regression analysis. By using this approach, points that negatively impact the regression model can be detected. Cook's distance is related to the leverage and residuals of observations, greater the leverage and residual value greater the cook's distance. If Cook's distance D_i value is more than 0.5 the ith data set needs to be additionally investigated and if D_i value is exceeding 1 then the ith data set is considered as influential. After the removal of ith data point, the cook's distance also measures the changes in the model's fitted value. Any area with Cook's Distance larger than 4/n, where n represents the sum of data points, is regarded as an outlier. It is essential to remember that a cook's distance is frequently applied as a technique to highlight key information. An influential data point need not necessarily to be eliminated just because it is influential. Fig. 1 displays the influence on the y-axis, instance index on the x-axis, and the dashed line, which represents the influence (threshold value) of the point. Outliers can be distinguished from other points by having a large residual (often larger than or equal to $\pm$ 2), but not all cases with a large residual are outliers. The potential for a case to impact the model is known as leverage. High leverage points are the farthest to the right. If a point fits the broader model without that situation, it may not have much of an impact on the model. Cook's distance outliers' detection graph shows that the dotted lines represent the limit while the blue vertical lines are the outliers. The longest outlier close to instance index 250 and influence up to 0.35 affects machine learning model the most.

About 3000 descriptors (features) are screened by applying Dragon software [47]. Descriptors are typically vector, matrix, integers, or another (special characters) data structures, which can be utilized for predicting molecule's property in a machine learning process. Descriptors can be categorized into both experimental descriptors and theoretical descriptors. The theoretical descriptors (e.g., empirical formula, structural formula etc.) could be further classified into five classes. 0D descriptors are called count descriptors as they provide information about atoms involved and molecular weight rather than structure and connectivity. 1D descriptor contains a list of structural fragments such as functional groups. Common 1D descriptors are fingerprints. 2D descriptors provide information based on graphical representation, and bonding of atoms. 3D descriptors are called the geometric descriptors that explain about size, surface, and spatial co-ordinates of atoms in molecules such as 3D-MoRSE, WHIM, GETAWAY descriptors and distinguish between isomeric compounds. 4D descriptors are also known as grid-based descriptors. These descriptions add a fourth dimension on top of the molecular geometry. This extra dimension typically describes interactions between the molecule and the receptor's active site or the molecule's many conformational states. CoMFA and GRID are common 4D descriptors. The 4D descriptors offer more information than the other descriptors and always produce varied values for molecules with various structural characteristics.

Molecules properties can be predicted by using descriptors in machine learning approaches [48,49]. The selection of a descriptor is

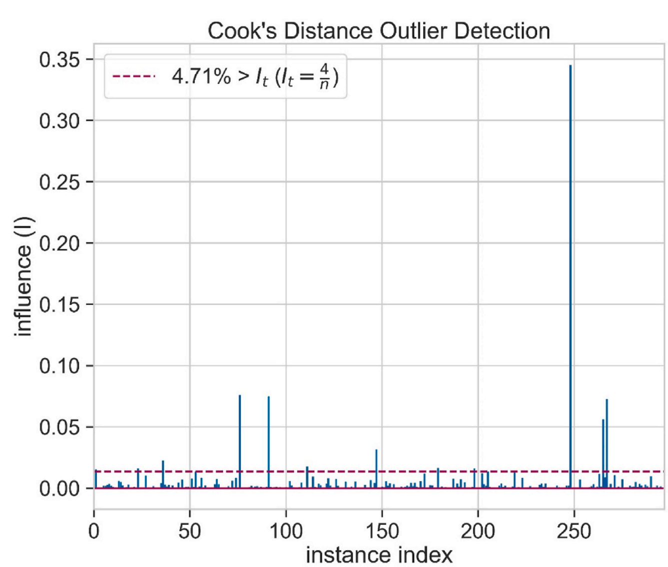


Fig. 1. Outlier in dataset.

crucial since it has a significant impact on how well a model predicts outcomes. The complexity of a molecule's structural makeup is beyond the scope of descriptors. Using several descriptors or high-dimensional descriptors can help to partially alleviate this problem [50]. However, it should be remembered that employing high-dimensional descriptors or many descriptors makes the task more complicated and therefore has a negative impact on how well the machine learning algorithm performs. Various libraries based on python such as sci-kit learn, seaborn, matplotlib and others are applied for visualization and machine learning analysis. The overview of ML process used in this study is shown in Scheme 1.

2.2. Similarity analysis

Similarity analysis is performed by using a cheminformatics tool known as RDKit [51]. Similarity analysis is a straightforward and convenient method to determine resemblance/similarity between reference structure and the structures that exist in database(s). To estimate the Tanimoto similarity, Extended Connectivity Fingerprints (ECFP4) are used. It is a unique way of performing similarity of chemical structures by comparing their fingerprints.

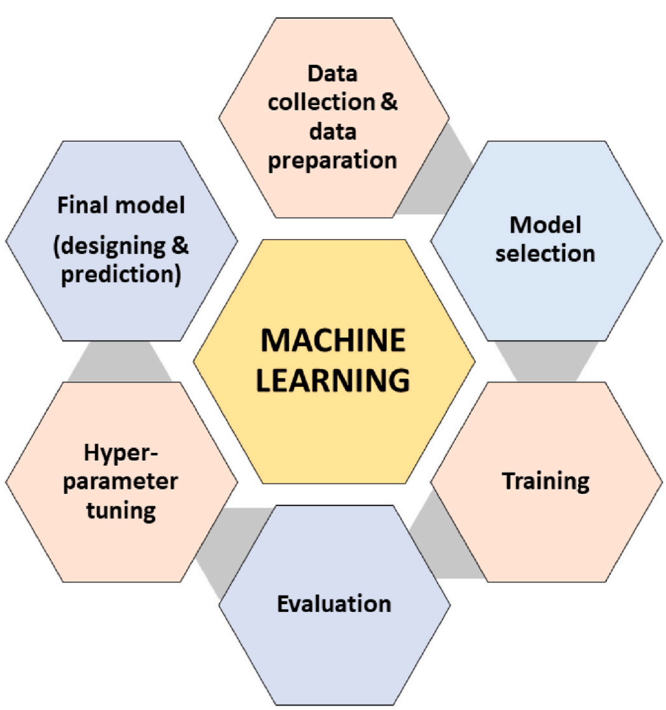
2.3. Library generation

Electron-deficient and electron-rich organic fragment's library is generated by using Data Warrior software [52]. To achieve this target, machine learning algorithms are employed. Through structural modification of reference structure, this approach can help in creating a variety of new molecules. In this way, it can generate a fitness score for each newly generated molecule. The molecules having more structural similarity with reference molecules assigned a good fitness score and can be used in the designing of new semiconductors.

3. Results and discussion

3.1. t-distributed stochastic neighbor embedding (t-SNE)

t-SNE is a statistical and unsupervised approach, which can be used for visualization of data by giving location to each data point on a 2D or 3D graph to visualize high-dimensional data. t-distribution is used to determine how similar two points in lower-dimensional space are while keeping the data's local structure. *t*-SNE is utilized for visualization and exploration of data in various fields, such as genetics, speech recognition, music interpretation, drug development, bioinformatics, geoscience domain interpretation, and biomedical data processing. Fig. 2shows the *t*-SNE plot for descriptors. Each dot represents a single descriptor. The data points are dispersed between the range of 1.50–3.25. In *t*-SNE plot, the location of each data point depends on the similarity of one structure with others. The compounds having similar structures grouped in an individual location on the *t*-SNE plot, whereas compounds having different structures will stay apart on the *t*-SNE plot.



Scheme 1. The overview of machine learning process used in present work.

3.2. Machine learning models

As no single model can make accurate predictions, more than one ML model is tested. Machine learning methods known as "many-model" use various models for collective learning. This approach of collective learning of many models is used to solve complex problems. For the selection of suitable machine learning models, normality information is useful. Some models need typical information such as gaussian naive bayes, logistic regression, linear regression etc. Different ML models are tested while extra trees regressor and CatBoost regressor give better results. Extra trees regressor are selected for further analysis. In machine learning, MAE (mean absolute error) is used for measuring performance along with the regression models (Table 1). Absolute error refers to the extent of divergence between an observation predicted value and true value. The MAE value needs to be as near to zero as possible. The MAE value is closer to zero the better the result, the lower the MAE more successfully a model matches a dataset. As shown in Table 1, most machine learning models reveal value closer to 0.1, which proves that these models can give better/accurate performance. Extra trees regressor has MAE value of 0.101 that is closer to zero showing better performance and accuracy of results as compared to CatBoost Regressor 0.107, Extreme Gradient Boosting 0.106, Gradient Boosting Regressor 0.174 whose values are far from 0. In addition, Passive Aggressive Regressor having MAE value 0.540 is less efficient in performance.

R square (R^2), coefficient of determination shows how fit a model is for targeted property. For this purpose, we firstly train the machine learning models along with linear regression algorithm and then its R^2 is determined. The value of R^2 close to 1 show that correlation between two variables is accurate. Some models have stronger prediction capability than others such as Extra Trees Regressor with R^2 of 0.595.

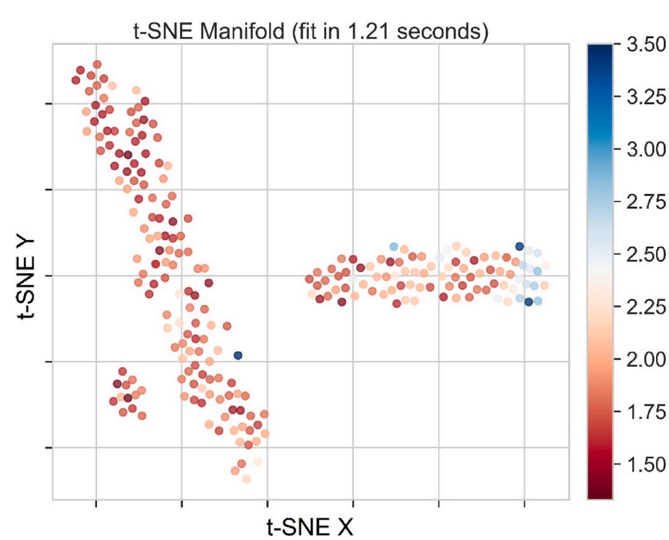


Fig. 2. *t*-SNE plot of descriptors.

Table 1
Performance (R^2 for test set; MAE) of different ML models.

Models	MAE	R^2
Extra Trees Regressor	0.101	0.595
CatBoost Regressor	0.107	0.584
Extreme Gradient Boosting	0.106	0.574
Gradient Boosting Regressor	0.174	0.469
Random Forest	0.172	0.467
AdaBoost Regressor	0.186	0.414
Light Gradient Boosting Machine	0.187	0.362
Ridge Regression	0.192	0.307
Bayesian Ridge	0.195	0.298
Random Sample Consensus	0.189	0.292
Linear Regression	0.194	0.288
Least Angle Regression	0.195	0.287
TheilSen Regressor	0.190	0.287
Support Vector Machine	0.199	0.280
K Neighbors Regressor	0.193	0.262
Lasso Regression	0.205	0.236
Elastic Net	0.205	0.234
Orthogonal Matching Pursuit	0.209	0.197
Lasso Least Angle Regression	0.246	−0.031
Decision Tree	0.225	−0.102
Huber Regressor	0.251	−0.224
Passive Aggressive Regressor	0.540	−5.302

Tables 1 and 2 shows MAE, R^2 values and RF for 10-fold cross-validation of different ML models, respectively.

3.2.1. Tuning hyperparameters

Tuning hyperparameters is crucial to control machine learning model's behavioral pattern for optimal performance. A hyperparameter is used in machine learning model training process to maximize its performance and minimize errors. A group of unique records (instances) containing the crucial properties of machine learning problems create input data, also known as training data. Using this information, the machine learning model can be trained for making precise predictions about targeted property. Hyperparameters tuning involves steps to train machine learning models, assessing the overall accuracy, and adjusting it for optimal performance.

3.3. Feature importance

Feature importance refers to assigning a value to each selected feature (descriptor) in each model, indicating the "importance" of each feature. Features are important for machine learning model prediction. The values assigned to each descriptor determine its contribution towards the model. It describes feature importance using a numeric value/score, wherein, the greater the score/value indicates more importance of that feature. By these numerical values irrelevant/less prominent features can be removed. All the descriptors do not have the same performance during ML training process. As shown in Fig. 3, EE_Dzs has a high variable importance greater than 0.15, while nT5HeteroRing had low variable importance between 0.04 and 0.05. However, decreasing or increasing the numbers of features significantly impacts the performance of ML model.

Table 2
10-fold cross-validation for Extra Trees Regressor.

	MAE	R^2
0	0.125	0.613
1	0.106	0.697
2	0.108	0.603
3	0.153	0.592
4	0.103	0.697
5	0.120	0.689
6	0.130	0.606
7	0.126	0.676
8	0.136	0.595
9	0.140	0.574
Mean	0.125	0.636
SD	0.015	0.046

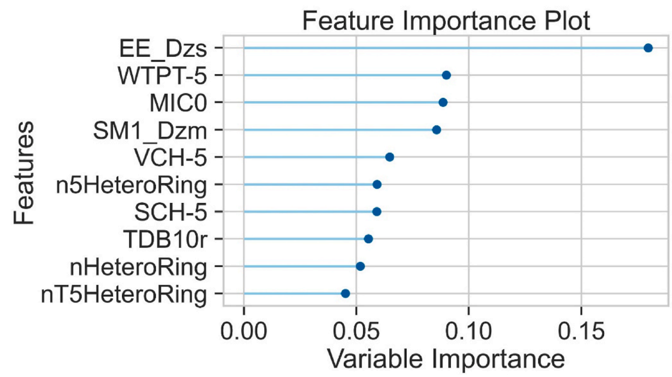


Fig. 3. The relative importance of features.

3.4. Learning curve for extra trees regressor

Learning curves are commonly utilized as a diagnostic tool in ML for algorithms, which learn incrementally (e.g., deep learning). This plot can also be used to compare the performance of ML models on testing and training data over changing training instances. It can also be used to diagnose overfit or underfit models during training time. Extra trees, an ensemble ML algorithm, which combines the predictions of several decision trees. Many features can be selected for all trees. It works by selecting a subset of features at random and then applying these for training a decision tree. Afterwards, the tree retains the highly significant traits for prediction. Extra trees have outperformed other algorithms like random forests. It has many advantages such as reducing bias, avoiding duplicate samples, and performing targeted property in a rapid and computationally low-cost manner.

Fig. 4 displays the learning curve for extra trees regressor i.e., consisting of training and cross validation scores to evaluate performance of ML model. As the number of training instances rises, the difference between training and cross validation scores tends to narrow down. If the two curves (scores) intersect, it indicates that the used data is not good to train ML model. The obtained results reveal that the two scores do not intersect, indicating the used data is useful to train ML model. To

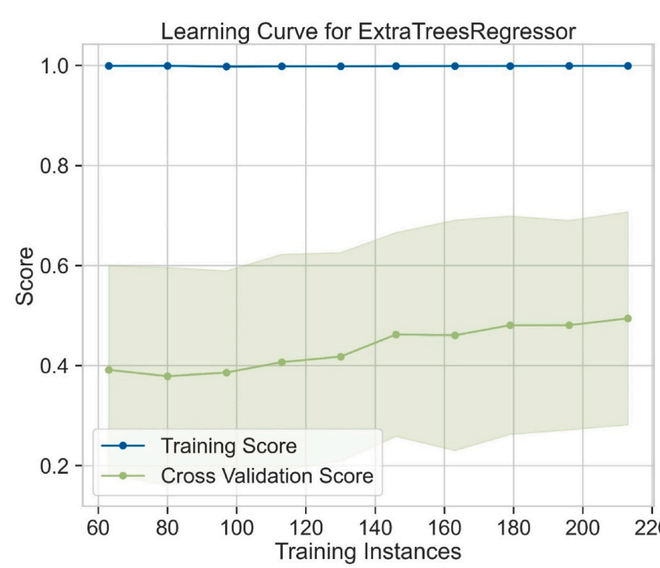


Fig. 4. Learning curve for extra trees regressor model.

improve prediction accuracy, more training instances and additional information (features) can be introduced.

3.5. Validation curve for extra trees regressor

Validation curve (plot) is used to select the most appropriate ML model (hyper)parameters. In contrast to learning curve, it can aid in assessing the model overfitting vs underfitting problems (bias-variance issue) against the model parameters. Fig. 5 displays the validation curve of extra trees regressor. The score is plotted over a changing hypermeter on the validation curve. To identify whether the estimator is overfitting or underfitting for certain hypermeter values, it is simpler to plot the impact of a particular hypermeter (max_depth) on the training and validation score. The depth of the ML model reveals that the model has more (stronger) capability to achieve added information.

3.6. Residuals for ExtraTreesRegressor

A residual plot graphically shows the relationship between train and test variables. A regression line's residual value is an indicator of how effectively the dataset is modelled by revealing data points fitting and missing data points. Fig. 6 shows that residuals are plotted on the y-axis, while the predicted values are displayed on the x-axis. The appropriateness of the ML model is indicated by the R^2 values. The plot shows that more data points are close (above and below) the fitting line, indicating the reasonable prediction capability of extra trees regressor.

3.7. Prediction error for ExtraTreesRegressor

Prediction error is used to indicate the difference between ML-based predicted and actual values. Fig. 7 shows a prediction error plot that is used to reveal how well the extra trees regressor predicts the targeted variables. In the plot, y indicates the observed value and y-hat symbolizes the predicted values, respectively. It can be used to locate the area with maximum errors. The obtained results revealed that most of the data points are dispersed around the best fit line, proving reasonable accuracy (low prediction error) of the extra trees regressor.

3.8. Similarity analysis

Similarity analysis is used to identify the similarities among molecules (reference molecules and molecules present in database). RDKit is

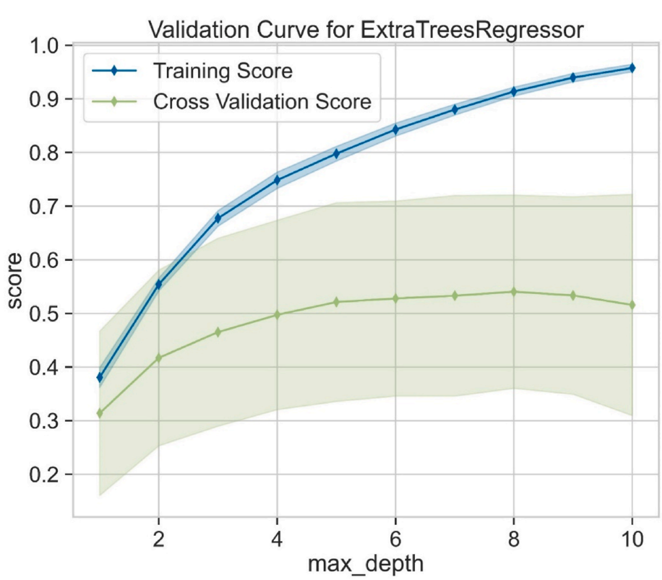


Fig. 5. Validation curve for extra trees regressor.

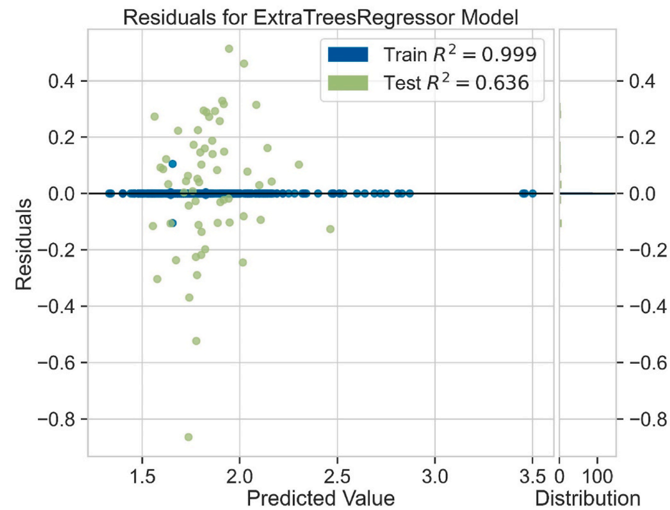


Fig. 6. Residuals for ExtraTreesRegressor.

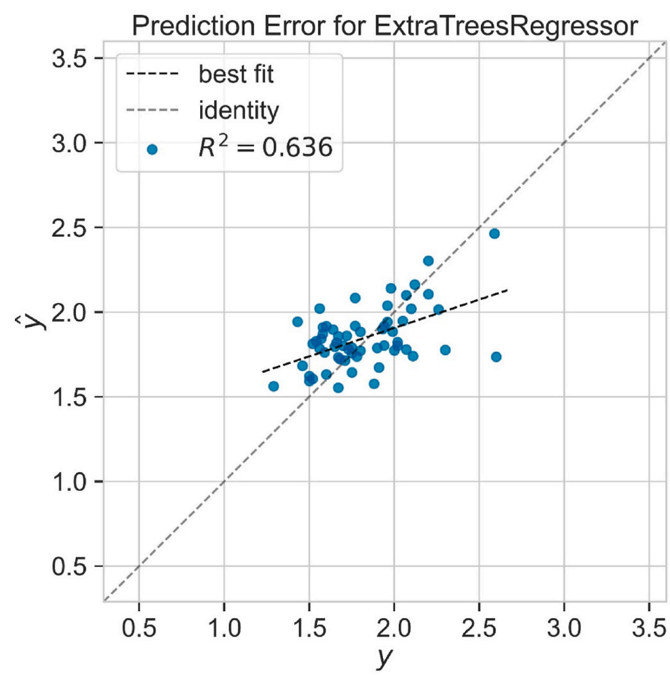
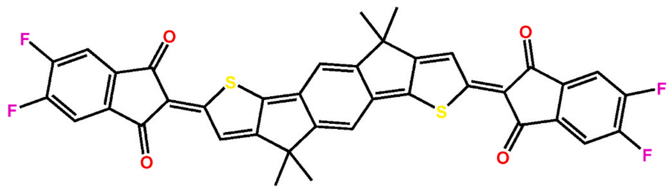


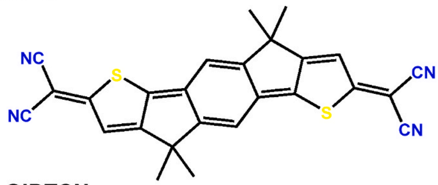
Fig. 7. Prediction error plot.

employed to perform similarity analysis [51]. The purpose is to check whether the molecules are similar or different from one another [53,54]. It involves determining similarities in chemical structures among molecules, which may show comparable properties. Structure-based similarity analysis is a crucial phenomenon for inventing new materials [55–57]. The structure of reference organic semiconductors used for similarity analysis given in Fig. 8. These semiconductors can be effectively utilized in thin-film transistors and in OSCs. Two low band gap organic semiconductors named QIDT-4F and QIDTCN are selected as reference structures [58]. These are famous and excellent organic semiconductors. The top search hits (with Tanimoto index) for QIDT-4F, QIDTCN, from Harvard clean energy project (CEP) database and QIDT-4F, QIDTCN from GDB17 chemical database are shown in Figs. 9–12, respectively.

Using the high technological CEP database, designing, and screening of advanced organic photovoltaic materials are achieved. CEP is used for the advancement of OPV for generating high performance compounds. It



QIDT-4F



QIDTCN

Fig. 8. Structure of reference organic semiconductors.

is a mechanized and highly efficient system for the in-silico investigation and study of molecular materials. It is composed and applied for the classification of organic materials for photovoltaic applications. For the investigation of millions of molecular chemical motifs and to determine their usage in electronic applications, CEP employs first-principles electronic structure theory by machine learning. For establishing a rich photovoltaic system, convenient molecular building blocks are needed. The Harvard organic database employed chemical similarity analysis to investigate comparable elementary units and elaborate structure and chemical composition relationship among the molecules. CEP is a significantly advanced project that provides use of the immense processing power of IBM's Global Community Grid.

CEP performs chemical similarity analysis using RDKit. In Fig. 9, QDIT-4F is used as a reference low band gap organic semiconductor compared with other molecules. The end-to-end benzene rings in the structure occupied with fluorine atoms exhibit a low band gap. The molecules having value close to 1 and chemical structure similar to the reference molecule may behave in the same manner both for properties and band gap. Among searched monomers, some can be utilized to create minor molecule acceptor. In Fig. 10, QIDTCN containing CN group on the terminals in the structure is taken as reference organic semiconductor. The results show that molecules with similar chemical structure can be used for synthetic purposes.

GDB17 contains billions of compounds having a variety of atoms (e.g., halogens, S, O, N, C and so on). GDB17 compounds are much more complex to synthesize. The exponential growth in the number of potential compounds with larger molecules and more complex structural elements, such as stereo centers, and heteroatoms, are one of the unique features of GDB database. In Figs. 11 and 12, QIDT-4F and QIDTCN are taken as reference structures, which are compared with compounds from the GDB17 database to find organic compounds with similar structures. In Fig. 11, structures with values 0.235, 0.224 are closer to

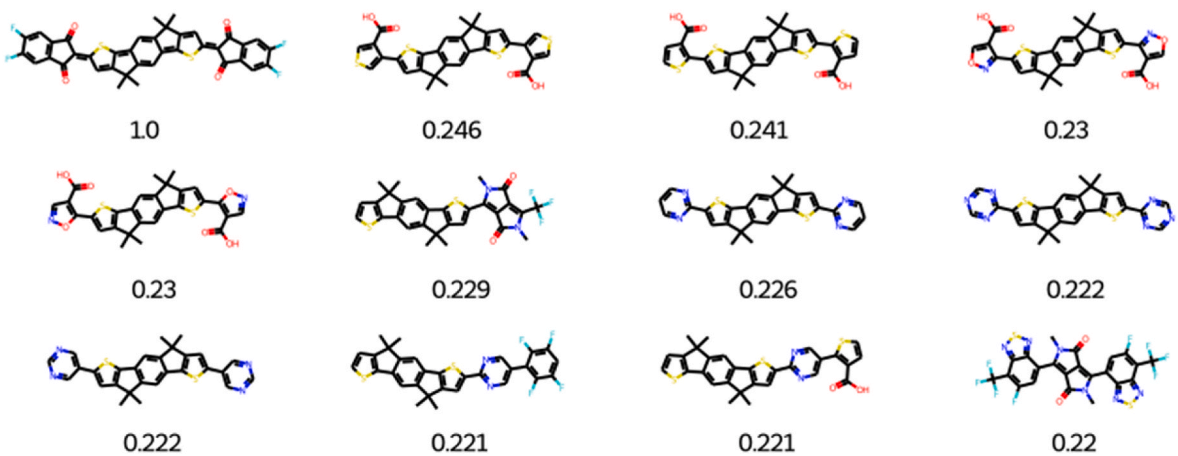


Fig. 9. Tanimoto index and top search hits for QIDT-4F using CEP.

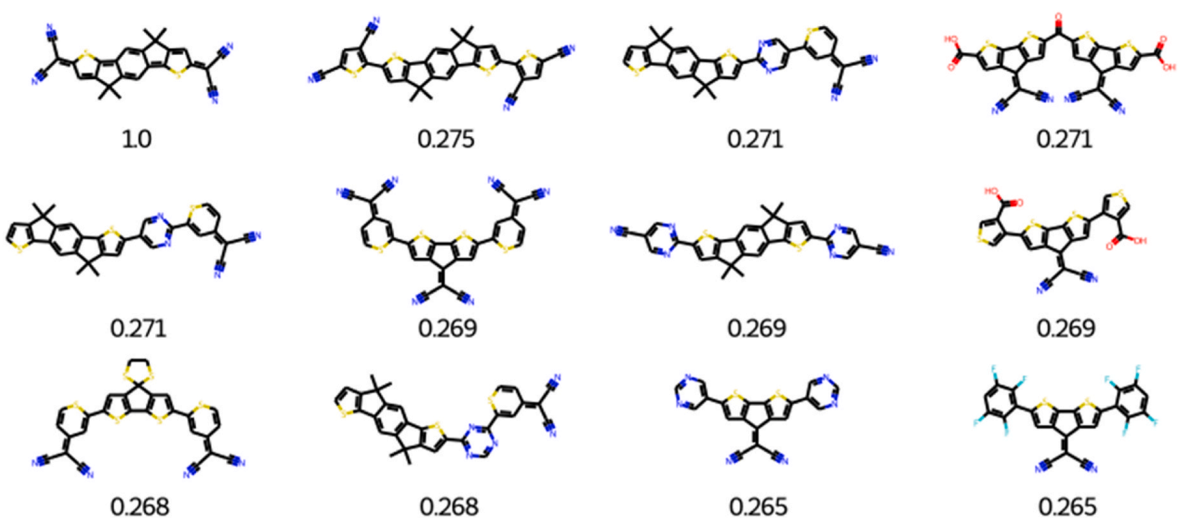


Fig. 10. Tanimoto index and top search hits for QIDTCN from CEP.

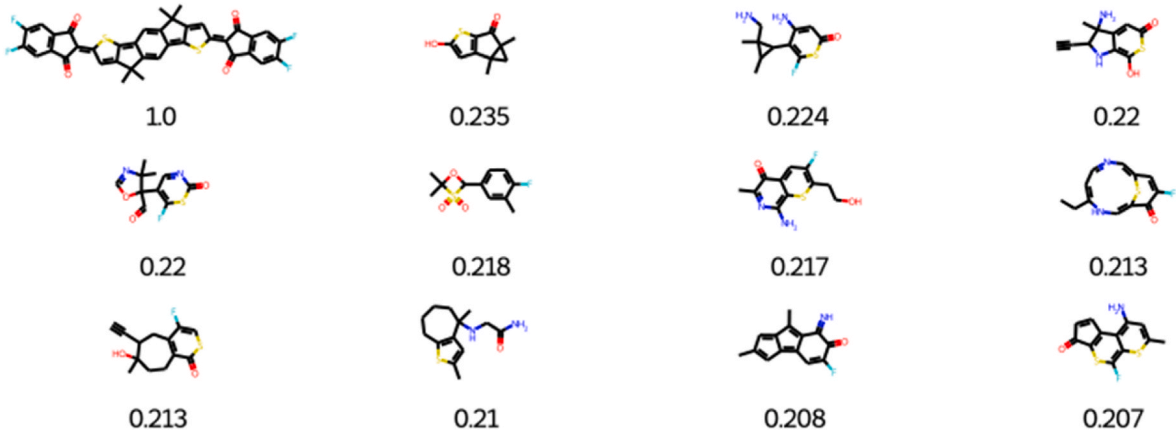


Fig. 11. Tanimoto index and top search hits for QIDT-4F using GDB17 database.

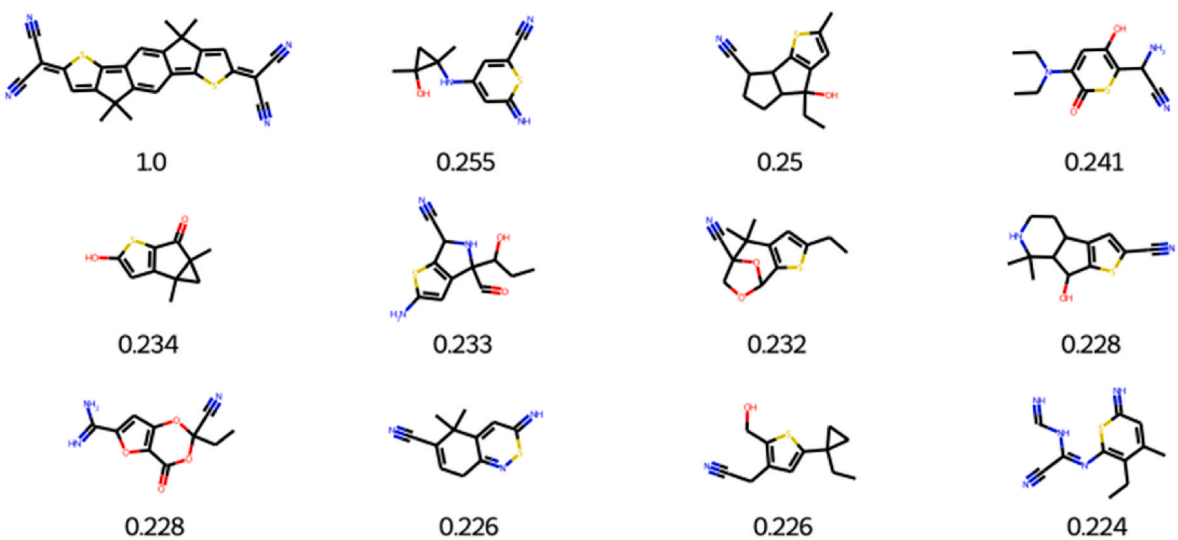


Fig. 12. Tanimoto index and top search hits for QIDTCN using GDB17 database.

the reference value and can show similar behavior [59]. Fig. 12 shows the top search hit for QIDTCN from GDB17 chemical database. In Fig. 12, structures with values 0.255, 0.25, 0.241 show the closest values near to 1.0.

3.9. Library enumeration

Library enumeration is the method used for producing a wide variety of small molecules (monomers/building blocks) by data mining and manual inspection [60,61]. The goal is to assemble a group of molecules that reflect a certain property or performance metric. Without investigation, the model prediction of molecules for specific application is not possible. However, notable features can be identified to provide guidance for the selection process. Virtual Library enumeration involves the collection and connecting molecules screened in computer-based applications. Library enumeration improves the variety of compounds and results in abundant molecule discovery and advanced outcome. Compounds discovered must have uniqueness and can be convenient to synthesis. Moreover, library enumeration helps in library creation for complex and conventional chemical structures [62,63]. The goal of this procedure is to produce novel compounds that have features or substituents that differ from those of existent compounds while maintaining their structural similarity to those molecules [64,65]. For example, if an existing compound has been found to have an undesirable property, then it can be altered by changing one or more atoms in the structure (or removing them entirely). This change will result in a compound that still has the same basic structure but has different properties.

Fig. 13 shows the top search hits related to QIDT-4F as reference structure. To enumerate the library, Data Warrior software is used. In Fig. 13, some compounds exhibit fitness scores greater than 0.8 while other compounds show scores of 0.91, 0.97, 0.92 close to the reference molecule QIDT-4F (i.e., having value of 1.0). Fig. 14 displays the score of similar compounds to that of QIDTCN. Most groups received a score of 0.9. In Fig. 14, most of the compounds show 0.959, 0.952, 0.944 closer value to the QIDTCN reference structure value. The obtained results revealed that most of the searched compounds can be utilized for generating potential targets for OSCs.

4. Band gap prediction of designed organic semiconductors

To design novel organic semiconductors on a large scale, organic building blocks are used [66–69]. Organic building blocks contain properties of their parent organic molecules but have functions of their own. Their functionality makes them unique from their parent molecules. One way of designing organic semiconductors is by combining different organic building blocks having chemical and functional properties (such as low band gap) similar to reference molecules (QIDT-4F and QIDTCN). And another way is by adding organic building blocks of interest in the center or terminal ends of reference molecule to increase its efficiency and applications.

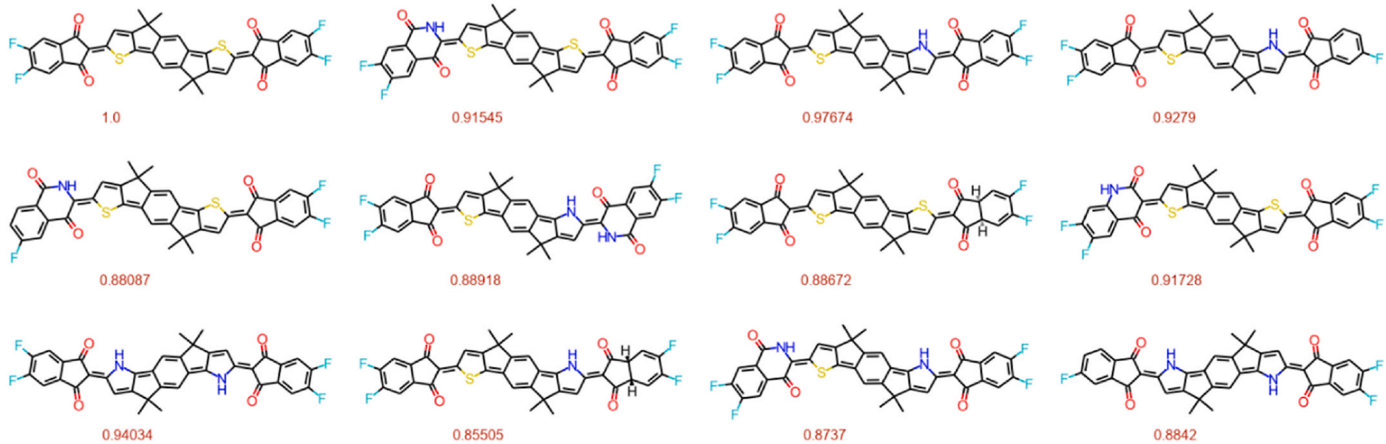


Fig. 13. Top 11 compounds similar to QIDT-4F (reference).

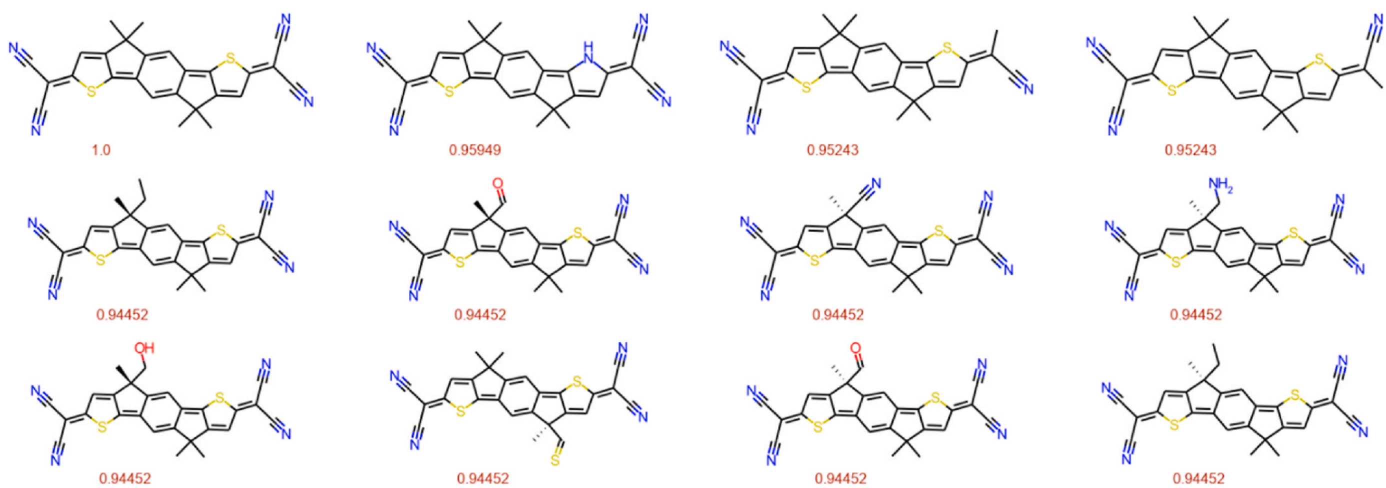


Fig. 14. Top 11 compounds similar to QIDTCN (reference).

Already generated organic fragments are being utilized to design novel organic semiconductors by using Breaking Retrosynthetically Interesting Chemical Substructures (BRICS) technique [70]. This technique contains different sized high quality organic fragments, that makes designing novel semiconductors easy and efficient. Extra trees regressor is an ML strategy used for regression and classification related issues. Gradient-boosted tree is the outcome of failure of decision tree. Gradient boosted trees can often defeat random forests. This model is built in a manner that it can operate for any distinct loss function. Extra trees regressor is used to predict the band gap of organic semiconductors. As stated earlier, semiconductors with the low band gap are preferred over semiconductors with the high band gap. Fig. 15 shows the band gap (eV) and chemical structures of the selected organic semiconductors.

5. Conclusions

In this study, multiple databases were mined to design low band gap organic semiconductors. By using obtained information, well-known and easy to synthesize organic groups were used for designing new molecules for semiconductors. Different machine learning models were trained and tested. Extra trees regressor was found to be the best fit model. Cook's distance was used to find out the negative observations. Similarity analysis was done to correlate database's molecules to reference molecules i.e., QIDT-4F and QIDTCN. Whereas library enumeration gave information about the building blocks of molecules. The newly designed low band gap semiconductors can help speed up the production of effective materials using a rapid and computationally low-cost approach. This methodology can aid in the synthetic approaches to avoid trail-and-error procedures and to increase the adoption of OSCs in the industrial setting.

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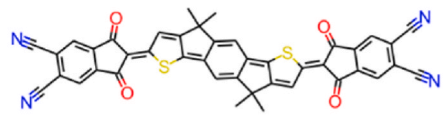
CRediT authorship contribution statement

Muhammad Saqib: Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Conceptualization. **Mashal Rani:** Writing – original draft, Visualization, Validation, Data curation. **Tayyaba Mubashir:** Writing – review & editing, Software, Formal analysis. **Mudassir Hussain Tahir:** Writing – review & editing, Visualization, Software. **Momina Maryam:** Writing – review & editing, Software, Formal analysis. **Afifa Mushtaq:** Writing – review & editing, Formal analysis. **Rafia Razzaq:** Visualization, Data curation. **Mohamed A. El-Sheikh:** Writing – review & editing, Visualization. **Hosam O. Elansary:** Methodology, Investigation, Funding acquisition.



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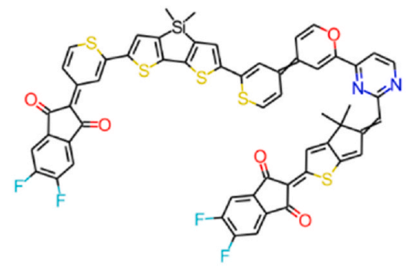

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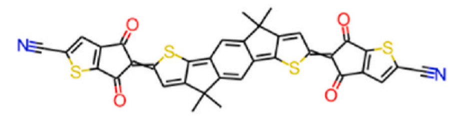

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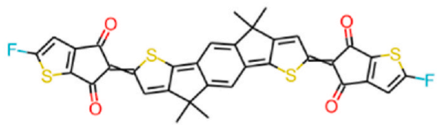

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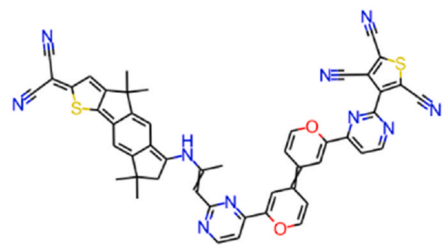

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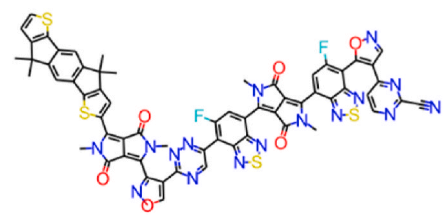

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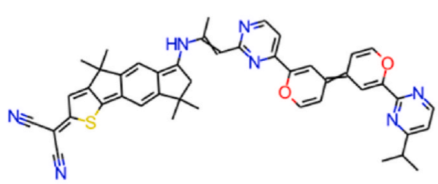

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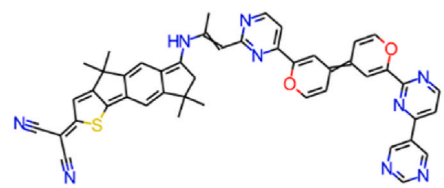

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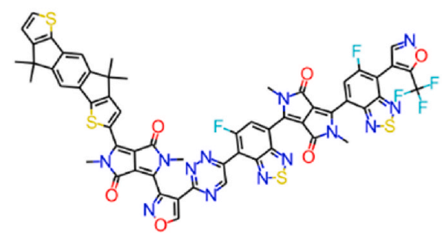

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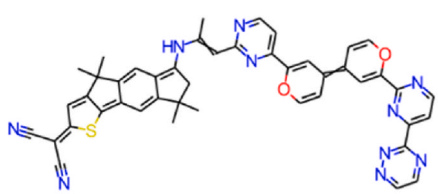

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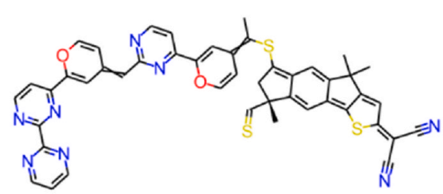

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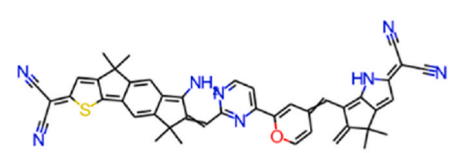

A12
1.415

Fig. 15. Chemical structures and band gap (eV) of selected organic semiconductors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.optmat.2024.115295.

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